Signals before action and experimental outcomes

20 performance trials | Three experiment blocks | Offline analysis of saved evidence

The twenty trials cover three separate experiment blocks. Eight twelve-partition trials at 700 messages/s provide the primary comparison; four four-partition trials at 700 and eight at 600 provide additional evidence. All actions were scheduled; these records do not demonstrate a learned or adaptive controller.

Redistribution confirmed the specified backlog-recovery criterion and completed every evaluation message in all six observed trials. This establishes observed sufficiency for those two criteria in the tested conditions, not for an unspecified application SLO. Its absence of observed failures prevents learning a redistribution success/failure boundary from these trials alone.

The twelve-partition block shows a trade-off: redistribution used 72 requested CPU core-minutes and had p99 of 80.10/84.16 s; Kafka scaling used about 137 core-minutes and had p99 of 71.41/72.46 s. Targeted scaling recovered sooner than redistribution in both repetitions, despite higher whole-cohort p99. Recovery and latency rank the responses differently.

The four-partition 700 block favors redistribution over targeted scaling on recorded p99, recovery time and requested resources in both repetitions. Keep 3 and native scaling were not tested in that block. The 600 block includes all four actions; Kafka scaling had the lowest p99 in Run 1 and redistribution in Run 2. Kafka scaling Run 2 completed during drain without confirming low-backlog recovery during input.

Starting ownership concentration is constant in all twenty trials, and persistent-hot count is constant within each block. Skew values mostly distinguish the four- and twelve-partition designs. A pooled correlation could therefore reflect the experimental design or changed deployment rather than a transferable decision rule.

The present records support descriptive comparisons and hypotheses. They do not support a defensible general-purpose classifier or neural controller. No predictive model, feature-importance ranking, significance test or pooled correlation is fitted in this analysis.

The analysis preserves the original outcome definitions. No new performance trials were run.

Starting conditions: one row per trial

B is consumer-position lag; Q is processing backlog. Means use seconds 30, 32, ..., 58. Q growth spans seconds 28-58. CPU and RSS sum the original three consumer processes.

Block	Response	Run	Mean B (offsets)	Q growth (offsets/s)	Mean skew	Persistent hot	CPU (cores)	RSS (MiB)
12hot-700	Keep 3	1	27,888	283.2	4.92	12	1.295	148.5
12hot-700	Keep 3	2	28,233	271.3	4.95	12	1.285	149.1
12hot-700	Redistribute 3	1	28,023	268.8	4.91	12	1.285	150.4
12hot-700	Redistribute 3	2	29,396	269.1	4.94	12	1.286	151.0
12hot-700	Targeted 6	1	28,168	265.6	4.93	12	1.282	149.9
12hot-700	Targeted 6	2	29,523	294.4	4.94	12	1.291	151.8
12hot-700	Kafka 6	1	28,167	287.7	4.93	12	1.290	148.8
12hot-700	Kafka 6	2	29,172	279.3	4.95	12	1.295	150.4
4hot-700	Redistribute 3	1	31,136	308.0	14.22	4	1.261	149.9
4hot-700	Redistribute 3	2	31,739	321.4	14.06	4	1.274	151.1
4hot-700	Targeted 6	1	31,302	318.6	14.20	4	1.272	149.3
4hot-700	Targeted 6	2	31,503	302.6	14.04	4	1.271	152.4
4hot-600	Keep 3	1	25,912	265.7	14.10	4	1.385	143.2
4hot-600	Keep 3	2	26,218	264.2	14.33	4	1.403	143.5
4hot-600	Redistribute 3	1	21,244	248.3	14.25	4	1.404	134.7
4hot-600	Redistribute 3	2	25,791	256.8	14.33	4	1.407	142.6
4hot-600	Targeted 6	1	26,487	249.9	14.15	4	1.396	143.9
4hot-600	Targeted 6	2	26,245	258.5	14.34	4	1.404	143.9
4hot-600	Kafka 6	1	26,176	249.4	14.15	4	1.399	142.3
4hot-600	Kafka 6	2	25,348	245.2	14.36	4	1.405	141.7

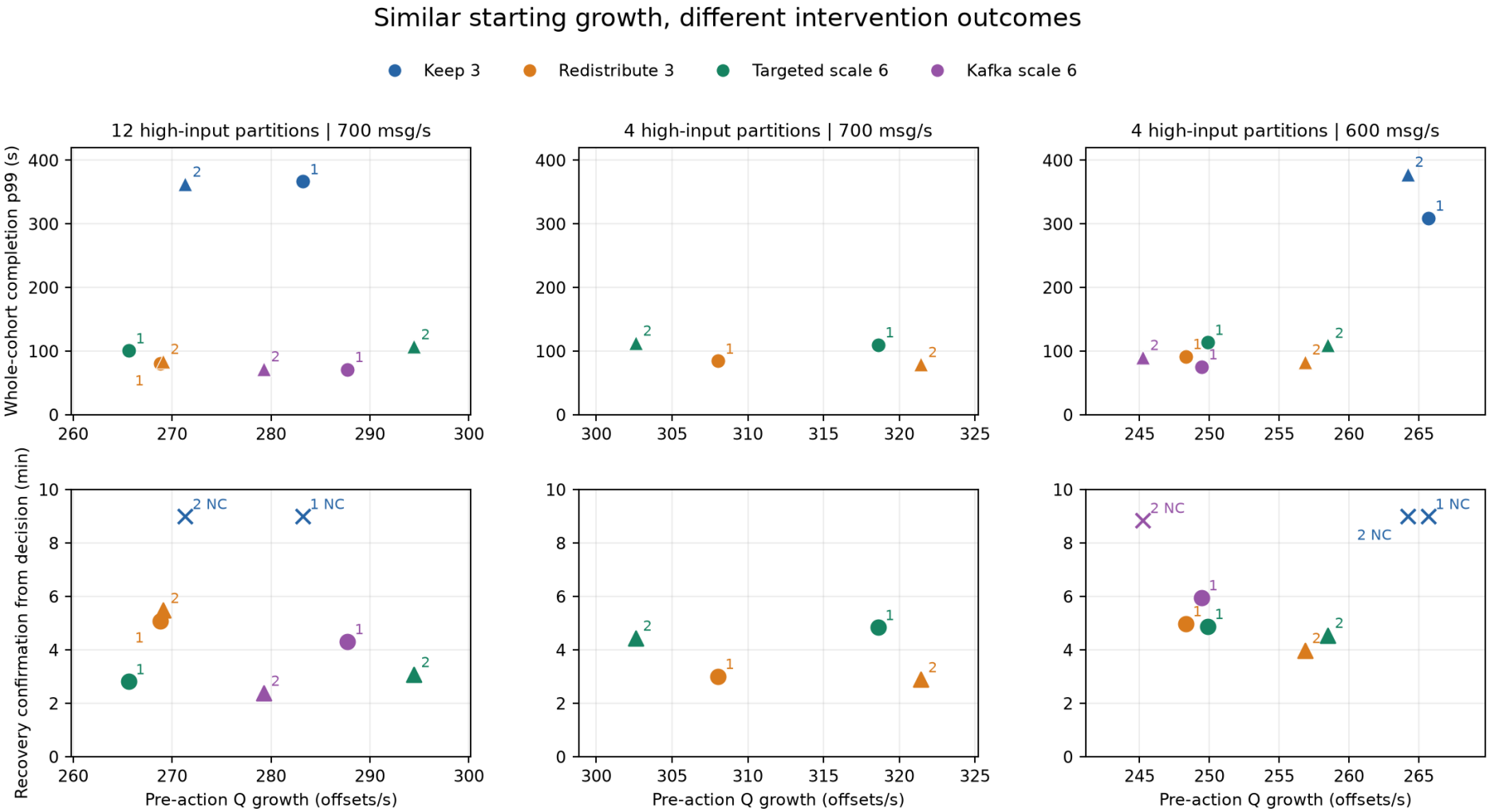
Performance, recovery and resource cost

p99 follows evaluation-born messages completed by the drain cutoff. Unfinished and one-second deadline outcomes use the full admitted cohort. Recovery requires Q <= 100 offsets for 30 valid continuous seconds during input.

Block	Response	Run	p99 (s)	Unfinished (%)	1 s misses (%)	Recovery (min)	Throughput (msg/s)	CPU requests (core-min)
12hot-700	Keep 3	1	366.64	34.64	83.32	Unconfirmed	423.2	Unavailable
12hot-700	Keep 3	2	362.79	34.83	83.36	Unconfirmed	422.6	72.0
12hot-700	Redistribute 3	1	80.10	0.00	30.50	5.09	727.7	72.0
12hot-700	Redistribute 3	2	84.16	0.00	31.84	5.48	729.5	72.0
12hot-700	Targeted 6	1	100.81	0.00	26.26	2.82	727.8	136.9
12hot-700	Targeted 6	2	107.24	0.00	27.33	3.08	729.3	136.8
12hot-700	Kafka 6	1	71.41	0.00	25.83	4.32	727.6	137.2
12hot-700	Kafka 6	2	72.46	0.00	21.05	2.38	728.5	137.1
4hot-700	Redistribute 3	1	84.50	0.00	26.63	3.01	730.9	72.0
4hot-700	Redistribute 3	2	79.28	0.00	25.82	2.89	731.4	72.0
4hot-700	Targeted 6	1	110.19	0.00	35.23	4.84	730.9	136.8
4hot-700	Targeted 6	2	112.94	0.00	34.72	4.45	731.1	136.8
4hot-600	Keep 3	1	309.29	26.92	85.66	Unconfirmed	392.2	72.0
4hot-600	Keep 3	2	378.35	37.57	85.70	Unconfirmed	347.1	72.0
4hot-600	Redistribute 3	1	91.82	0.00	38.76	4.99	617.6	72.0
4hot-600	Redistribute 3	2	83.29	0.00	31.32	3.98	625.2	72.0
4hot-600	Targeted 6	1	114.08	0.00	36.75	4.88	625.9	136.7
4hot-600	Targeted 6	2	109.42	0.00	33.43	4.55	625.5	136.5
4hot-600	Kafka 6	1	75.59	0.00	44.35	5.96	625.7	137.0
4hot-600	Kafka 6	2	90.28	0.00	66.75	Unconfirmed	597.4	136.9

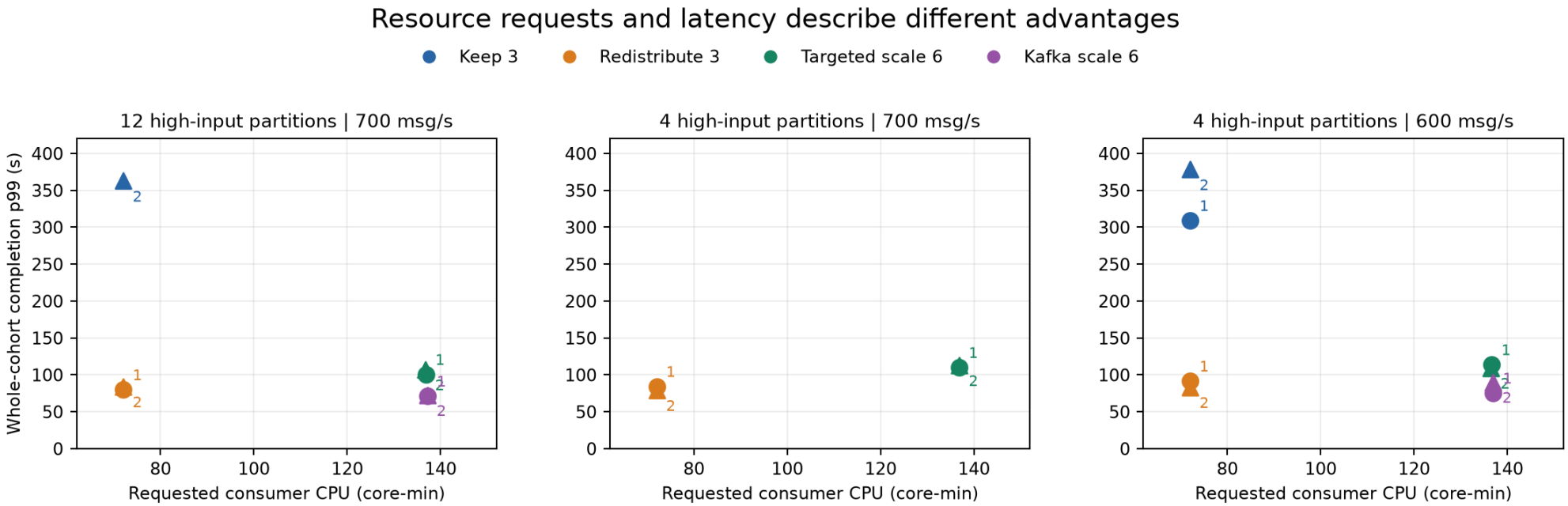
Throughput counts distinct completions during evaluation, including warm-up messages completed then, divided by 600 seconds. Unconfirmed recoveries are censored; they are not zero. The unavailable first Keep 3 request total is due to incomplete history. Cost covers evaluation plus drain and is distinct from actual CPU use. Application sufficiency remains unspecified.

Signals and outcomes within each campaign



One point per trial; labels identify Run 1 or 2. Growth: trailing 30 s ending at evaluation second 58.
NC: recovery unconfirmed; crosses show the observation cutoff, not a measured recovery time. Keep 3 uses second 60.
Cohort p99 excludes unfinished messages; their percentages are retained in the outcome table. Panels are separate campaigns.

Resource trade-offs and the next comparisons

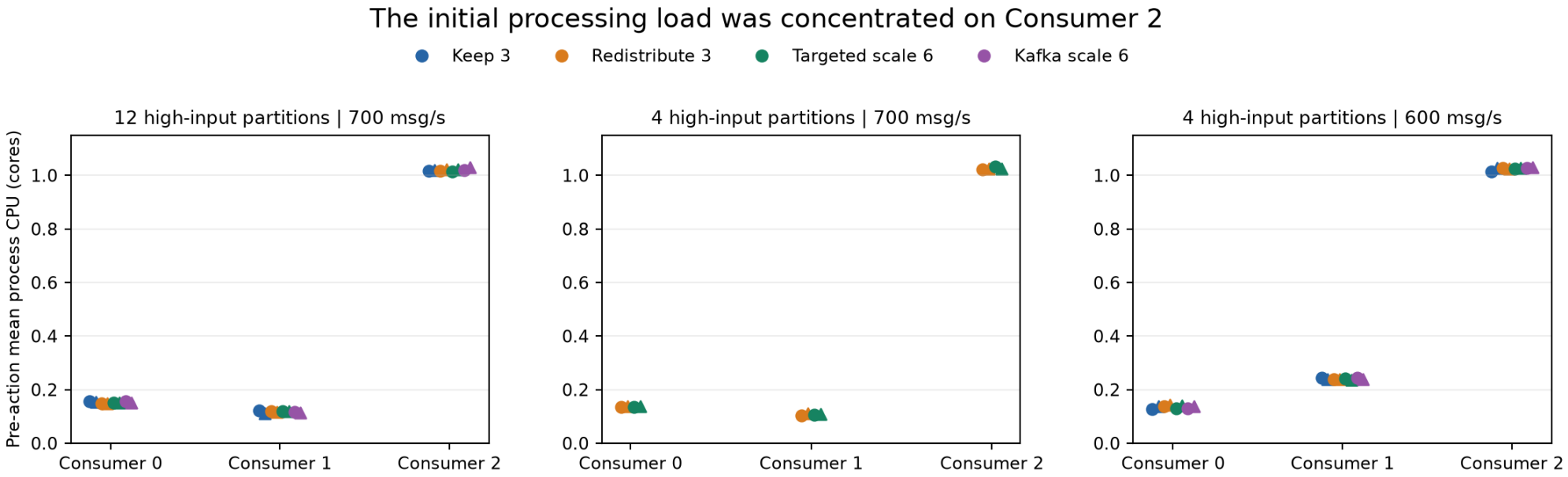


Requests cover evaluation plus drain. Keep 3 unfinished: 34.64/34.83% (12 hot, 700), 26.92/37.57% (4 hot, 600).
All intervention trials finished the evaluation cohort. Keep 3 Run 1 at 700 is omitted here because request coverage is incomplete.
No Keep 3 or native Kafka scaling trial exists in the four-partition 700 block. No cross-campaign percentage comparison is made.

First vary initial ownership at fixed workload and deployment: concentrated on one consumer, partly distributed, and already distributed. Compare the available actions in matched blocks with reversed or randomized action order. This directly tests whether ownership improves prediction beyond backlog and skew.

Then vary offered work relative to measured consumer capacity while keeping the ownership design controlled. Measure per-consumer service capacity under sustained available work; do not infer capacity from an underloaded completion rate or CPU percentage. Select load levels on both sides of the observed recovery boundary using calibration.

Consumer-level measurements and limits on learning



Arithmetic means at evaluation seconds 30, 32, ..., 58. Small horizontal offsets separate trials visually.
All selected high-input partitions initially belonged to Consumer 2. Low CPU on other consumers suggests uneven use,
but CPU observations do not measure how much additional work those consumers can sustainably process.

Retain unfavorable outcomes and unconfirmed recoveries. Keep trial/campaign/seed groups together in future validation; neighboring scrapes and windows are not independent examples. Reserve separate workload schedules for final testing and record deployment changes.

Specify acceptable deadline-miss, unfinished-work and recovery requirements before assigning an application-sufficiency label. Start with a simple interpretable baseline only when the experiments include diverse conditions and both sufficient and insufficient redistribution outcomes.

Reproducibility: analysis.json contains the definitions, source SHA-256 hashes, pre-action measurements and original outcomes. trial-data.csv provides one record per trial. All three blocks must remain identifiable in subsequent analysis.